

# A permutative cellular automaton with positive propagation speed

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**Disproof of conjecture 00000009623.** The conjecture asserts that a surjective cellular automaton has zero damage propagation speed if and only if its rule is a permutative factor. The binary coordinate shift refutes this clause.

Let  $X = \{0, 1\}^{\mathbb{Z}}$  have the product topology, with the finite alphabet discrete, and define  $F : X \rightarrow X$  by

$$(Fx)_i = x_{i+1}.$$

Each output coordinate is a coordinate projection, so  $F$  is continuous. It commutes with the one-dimensional shift, being that shift itself. Its inverse is  $(F^{-1}x)_i = x_{i-1}$ , so it is surjective. Its finite neighborhood is  $\{1\}$  and its local rule is  $f(b) = b$ . This rule is a permutation of the binary alphabet and hence is permutative. Even if a symmetric radius-one neighborhood is used, it is right-permutative: fixing the other coordinates, the map of the rightmost coordinate to the output is the identity permutation.

For every configuration  $x$ , let  $y$  differ from  $x$  only at site  $j$ , where its bit is flipped. Induction gives

$$(F^t x)_i = x_{i+t}, \quad (F^t y)_i \neq (F^t x)_i \iff i = j - t.$$

Thus the damage travels left by exactly  $t$  sites in  $t$  iterations. Its leftward spatial propagation speed is

$$\lim_{t \rightarrow \infty} \frac{|(j - t) - j|}{t} = 1.$$

This is spatial displacement, not the count of damaged sites, which remains one.

The same conclusion follows from the directional half-line front used for cellular-automaton Lyapunov exponents. Suppose  $x, y$  agree on  $[j, \infty) \cap \mathbb{Z}$ . Their iterates agree on  $[j - t, \infty) \cap \mathbb{Z}$ . This boundary is sharp: flip  $x$  at  $j - 1$  to produce  $y$ ; the disagreement is exactly at  $j - 1 - t$ . Hence the left-moving boundary advances exactly  $t$  sites. Reversing orientation swaps the two directional exponents; one is 1 and the other 0, so the propagation speed does not vanish.

These identities hold for *every* configuration, so they also hold almost surely under the fair Bernoulli product measure (indeed under any measure on  $X$ ). There is no exceptional set or probabilistic limit to estimate. This surjective permutative factor has positive propagation speed, contradicting the claimed equivalence. The other conjuncts need not be settled.

**Lean coverage.** The project formalizes actual bi-infinite Boolean configurations with the product topology, continuity, surjectivity, shift commutation, permutativity, the exact iteration formula, the unique damage site, and both the half-line bound and its sharpness for every configuration. The normalized displacement equals one at every positive time. Lean 4.19.0 and Mathlib are pinned; printed theorem audits contain only standard logical axioms.